

4 Technical Safety Concept „System Elements“

The safety requirements at system level shall be refined for each safety related system element (within the item)

Note: In the following chapters only the system element “Power Unit and Motor Monitoring” is considered. In real projects of course all involved system elements have to be considered as well.

4.1 Safety Requirements for the system element “Power Unit and Motor Monitoring”

This is a refinement of the safety requirements from system level to the system elements “Power Unit” and “Motor Monitoring”.

The references are given by the system safety requirements allocated to the considered system elements.

4.1.1 Safety Requirements to the detection and control of faults in the component

Ref.	ID	Requirement	ASIL
SYSR1	SYSELR1	The active motor current shall be measured continuously	B
SYSR1	SYSELR2	The value of the active motor current shall be sent to ECU continuously via analogue signal	B
SYSR6	SYSELR3		
SYSR6	SYSELR4		

4.1.2 Safety Requirements to the HW-Architectural Metrics at component level

The values are directly derived from system level.

Ref.	ID	Requirement	ASIL
SYSR10	SYSELR5		
SYSR11	SYSELR6		

4.1.3 Safety Requirements to the Probabilistic Metric for random Hardware Faults (PMHF value) at system element level

There is a budget given from the total PMFH target value

Ref.	ID	Requirement	ASIL
SYSR20	SYSELR7		